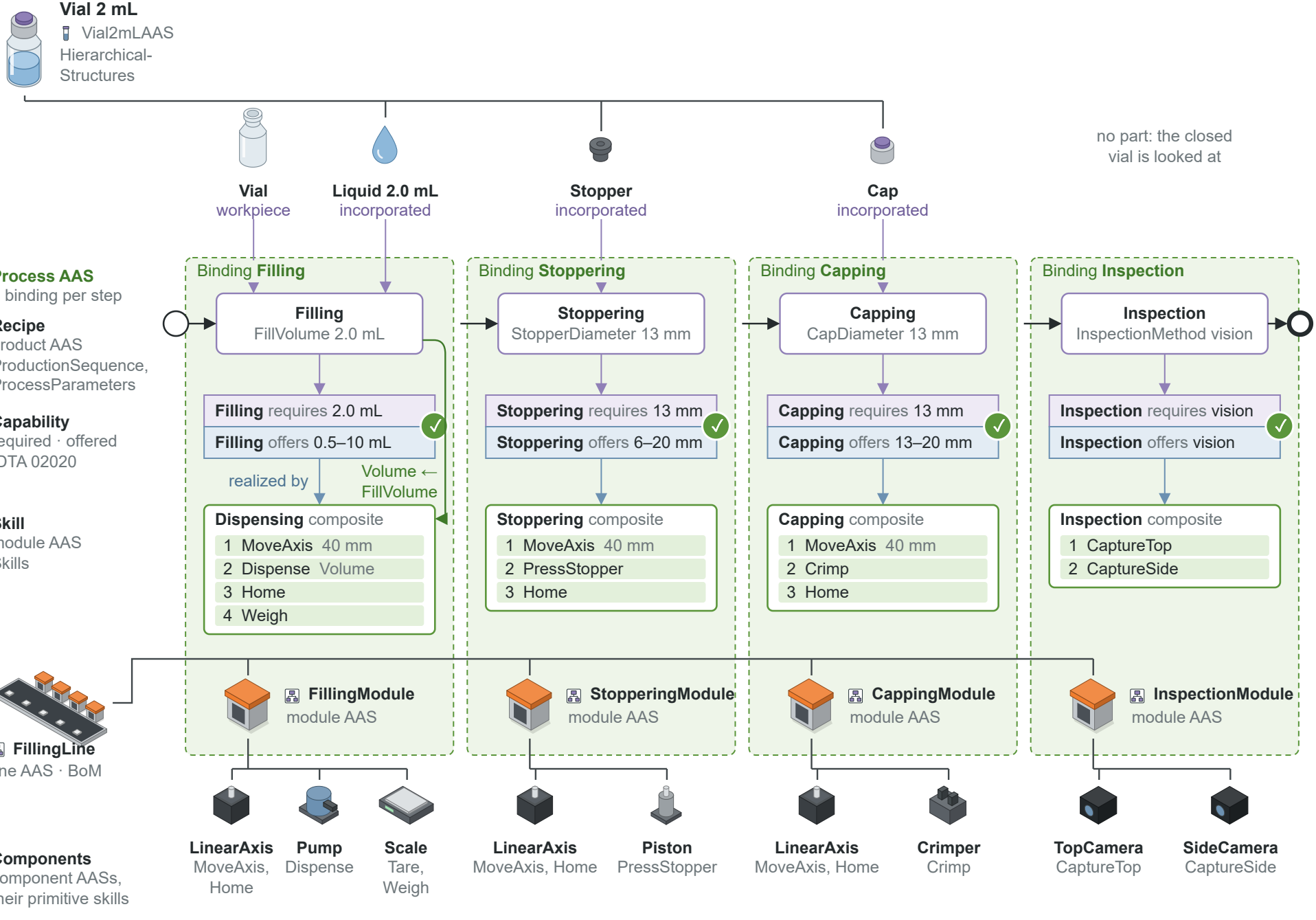

Vial 2 mL
Vial2mLAAS
Hierarchical-Structures
Vial
workpiece
Liquid 2.0 mL
incorporated
Stopper
incorporated
Cap
incorporated
no part: the closed vial is looked at
Process AAS
a binding per step
Recipe
product AAS
ProductionSequence, ProcessParameters
Capability
required · offered
IDTA 02020
Skill
module AAS
Skills
Components
component AASs, their primitive skills
Binding Filling
Filling
FillVolume 2.0 mL
Filling requires 2.0 mL
Filling offers 0.5–10 mL
realized by
Volume ← FillVolume
Dispensing composite
1 MoveAxis 40 mm
2 Dispense Volume
3 Home
4 Weigh
Binding Stoppering
Stoppering
StopperDiameter 13 mm
Stoppering requires 13 mm
Stoppering offers 6–20 mm
Stoppering composite
1 MoveAxis 40 mm
2 PressStopper
3 Home
Binding Capping
Capping
CapDiameter 13 mm
Capping requires 13 mm
Capping offers 13–20 mm
Capping composite
1 MoveAxis 40 mm
2 Crimp
3 Home
Binding Inspection
Inspection
InspectionMethod vision
Inspection requires vision
Inspection offers vision
Inspection composite
1 CaptureTop
2 CaptureSide
FillingLine
line AAS · BoM
FillingModule
module AAS
StopperingModule
module AAS
CappingModule
module AAS
InspectionModule
module AAS
LinearAxis
MoveAxis, Home
Pump
Dispense
Scale
Tare, Weigh
LinearAxis
MoveAxis, Home
Piston
PressStopper
LinearAxis
MoveAxis, Home
Crimper
Crimp
TopCamera
CaptureTop
SideCamera
CaptureSide